UML Example Exam Problem

1. Use the following requirements to complete various problems in this section

Requirements: KU Course Registration System

1. All users must log in with their KU account through KU Single Sign-On. User roles such as student, university admin, and instructor are provided by this system.	1. user ทุกคนต้อง log in ด้วย KU account ผ่าน KU Single Sign-On โดยที่ User roles เช่น student, university admin และ instructor จะมาจากระบบนี้
2. Department Admins can create new course information: course code, course name, description, credits, and fee.	2. Department Admins สามารถสร้างข้อมูล course ใหม่ได้: course code, course name, description, credits และ fee
3. Department Admins can set course offerings by first selecting a semester (e.g., 2025-1) and course code (e.g., CPE334). They then enter a sections list with details: section number, room, class time (e.g., “M W F 10:30–12:00”), display notes (e.g., ผู้สอน, หมายเหตุ), and maximum students allowed. For example, if CPE334 is offered in semester 2025-1 with sections 1, 2, 3, and 4, the system will create four course offerings for CPE334.	3. Department Admins สามารถกำหนด course offerings ได้โดยเลือก semester (เช่น 2025-1) และ course code (เช่น CPE334) ก่อน จากนั้นกรอก sections list พร้อมรายละเอียด: section number, room, class time (เช่น “M W F 10:30–12:00”), display notes (เช่น ผู้สอน, หมายเหตุ) และจำนวน maximum students allowed ตัวอย่างเช่น ถ้า CPE334 เปิดใน semester 2025-1 พร้อม sections 1, 2, 3 และ 4 ระบบจะสร้าง course offerings สำหรับ CPE334 ทั้งหมด 4 รายการ
4. University Admins can create new semesters (e.g., 2024-3, 2025-1, 2025-2) and set the currently active semester.	4. University Admins สามารถสร้าง semester ใหม่ (เช่น 2024-3, 2025-1, 2025-2) และกำหนด semester ที่เป็น currently active ได้
5. Students can open a Registration Form for the active semester. Initially, the form has no course offerings selected. The form automatically pulls data from the KU Student Info System using the logged-in username, displaying student ID, name, degree program, email, login username, etc.	5. Students สามารถเปิด Registration Form สำหรับ active semester ได้ โดยเริ่มแรก form จะยังไม่มี course offerings ถูกเลือก Form จะดึงข้อมูลจาก KU Student Info System โดยใช้ logged-in username เพื่อแสดงข้อมูล student ID, name, degree program, email, login username ฯลฯ
6. Students can add multiple course offerings to the Registration Form through the Course Offering List View. The total registration fee updates as courses are added.	6. Students สามารถเพิ่ม course offerings หลายรายการลงใน Registration Form ผ่าน Course Offering List View โดยที่ total registration fee จะอัปเดตตาม courses ที่ถูกเพิ่ม
7. The Course Offering List View displays: course ID, course code, course name, credits, fee, semester, section, room, class time, display notes, maximum students, number registered, and seats remaining. Students can filter and select courses to add to the registration form.	7. Course Offering List View จะแสดง: course ID, course code, course name, credits, fee, semester, section, room, class time, display notes, maximum students, number registered และ seats remaining โดย Students สามารถ filter และเลือก courses เพื่อเพิ่มลงใน registration form ได้

8. The Registration Form begins in Draft status. Students can save and later return to edit drafts before final submission.

9. Upon submission, the system checks seat availability for each course:
– If a section is full → submission fails, and a warning appears for students to correct and resubmit.
– If all sections have seats → proceed to payment.

10. Payment options include:
– Credit card via KBank payment gateway.
– Bank transfer via Bangkok Bank payment gateway.

11. After successful payment, the registration status becomes Done:
– Forms with Done cannot be edited online.
– Students receive a PDF copy of their registration by email via KU Email (Google Workspace).

8. Registration Form จะเริ่มต้นในสถานะ Draft โดย Students สามารถ save และกลับมา edit draft ก่อนการ submit ครั้งสุดท้ายได้

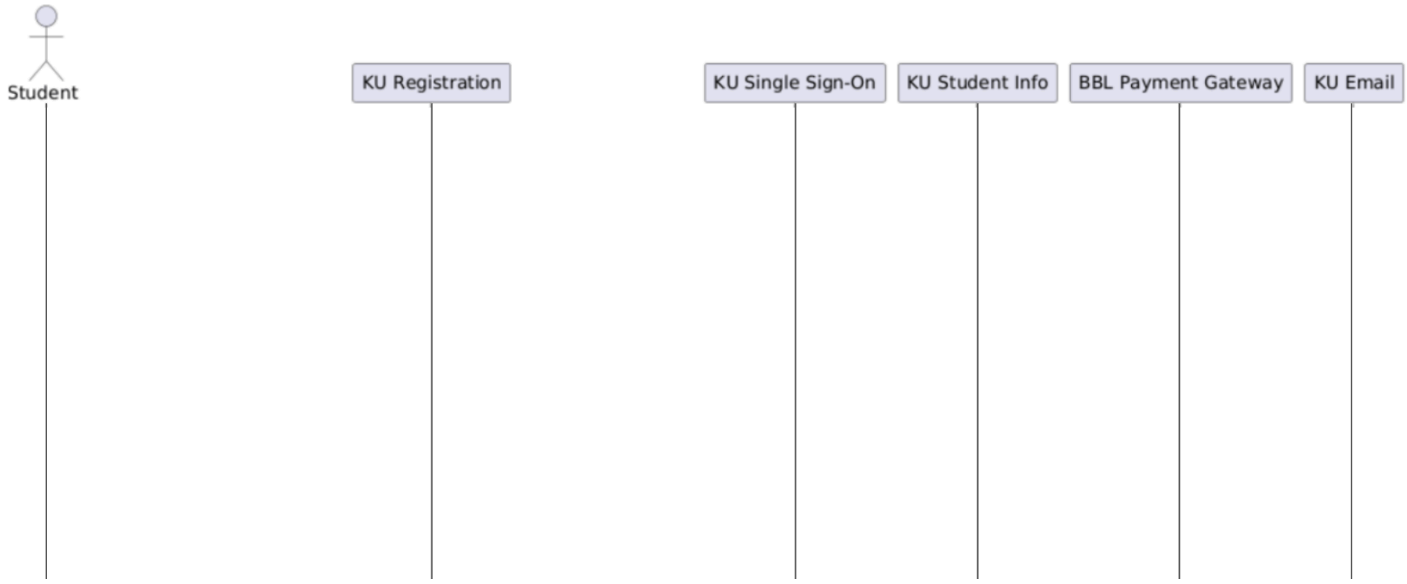
9. เมื่อ submit ระบบจะตรวจสอบ seat availability ของแต่ละ course:
– ถ้า section เต็ม → submission ล้มเหลว และจะมี warning ปรากฏเพื่อให้ Students แก้ไขและ resubmit
– ถ้าทุก section ยังมีที่นั่ง → ดำเนินการต่อไปที่ขั้นตอน payment

10. Payment options ได้แก่:
– Credit card ผ่าน KBank payment gateway
– Bank transfer ผ่าน Bangkok Bank payment gateway

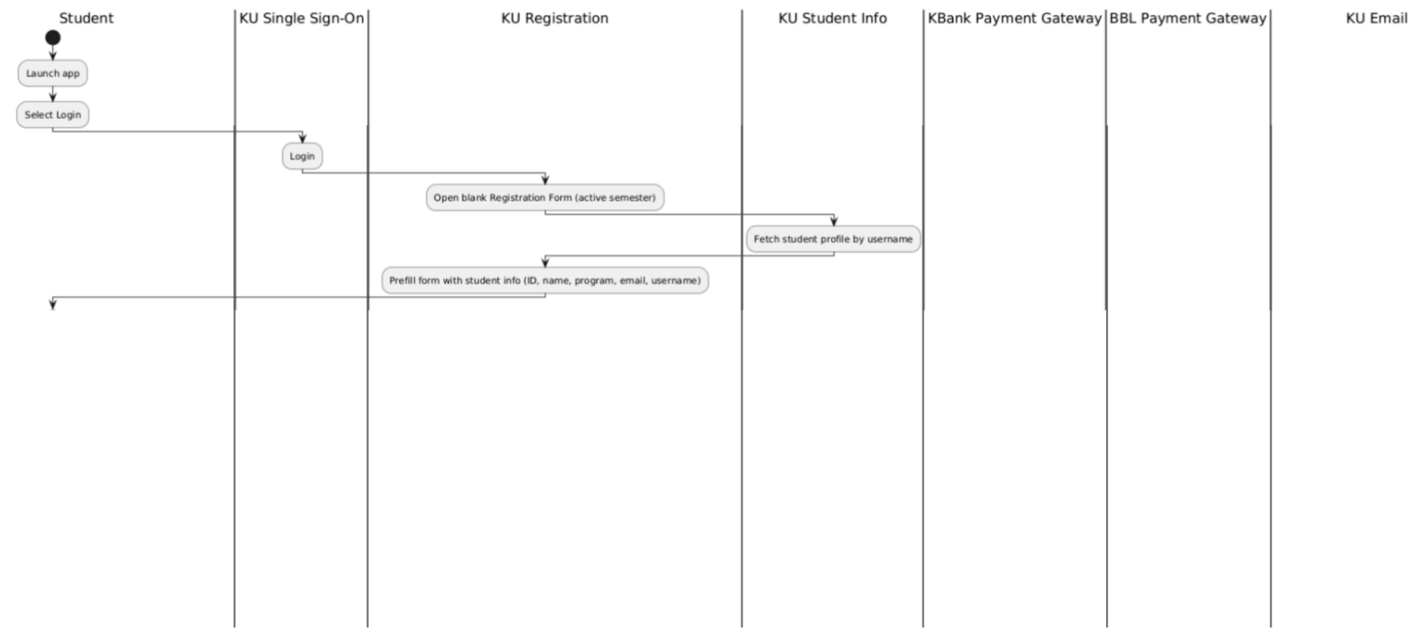
11. หลังจากการ payment สำเร็จ registration status จะกลายเป็น Done:
– Form ที่เป็น Done จะไม่สามารถ edit ออนไลน์ได้
– Students จะได้รับ PDF copy ของ registration ทาง email ผ่าน KU Email (Google Workspace)

1.1 *10 points*. **Complete this sequence diagram with swimlanes for the following end-to-end scenario:**

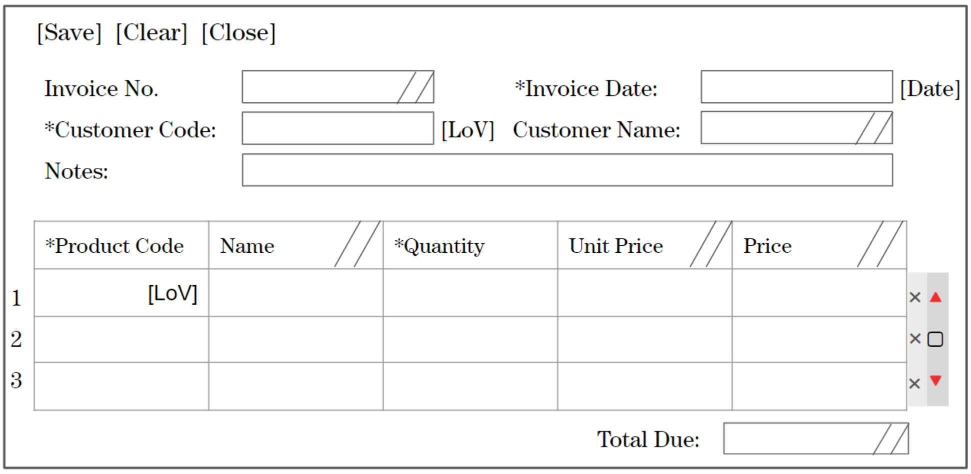
1. A student logs in.
2. The system detects that the student already has a Draft Registration Form for the current semester containing two course offerings, so the student is presented with that form to continue.
3. The student adds one more course offering.
4. The student submits the form, but one of the three course offerings is full so submission fails.
5. The student is informed.
6. The student changes that course offering to a different section and submits again.
7. This time the submission is successful.
8. The student pays by bank transfer.
9. The registration is completed successfully.



1.2 *10 points.* **Complete the UML Activity Diagram with partitions (swimlanes) shown below. It models the main registration workflow in which a student logs in for the first time this semester, receives a blank registration form containing their information, selects courses, pays successfully, and receives confirmation. Be sure to cover all decision branches (e.g., payment method) and loops (e.g., resubmit until confirmation is received) using simple flowchart notation for decision nodes. The workflow represents a single session and ends with the student successfully registered.**



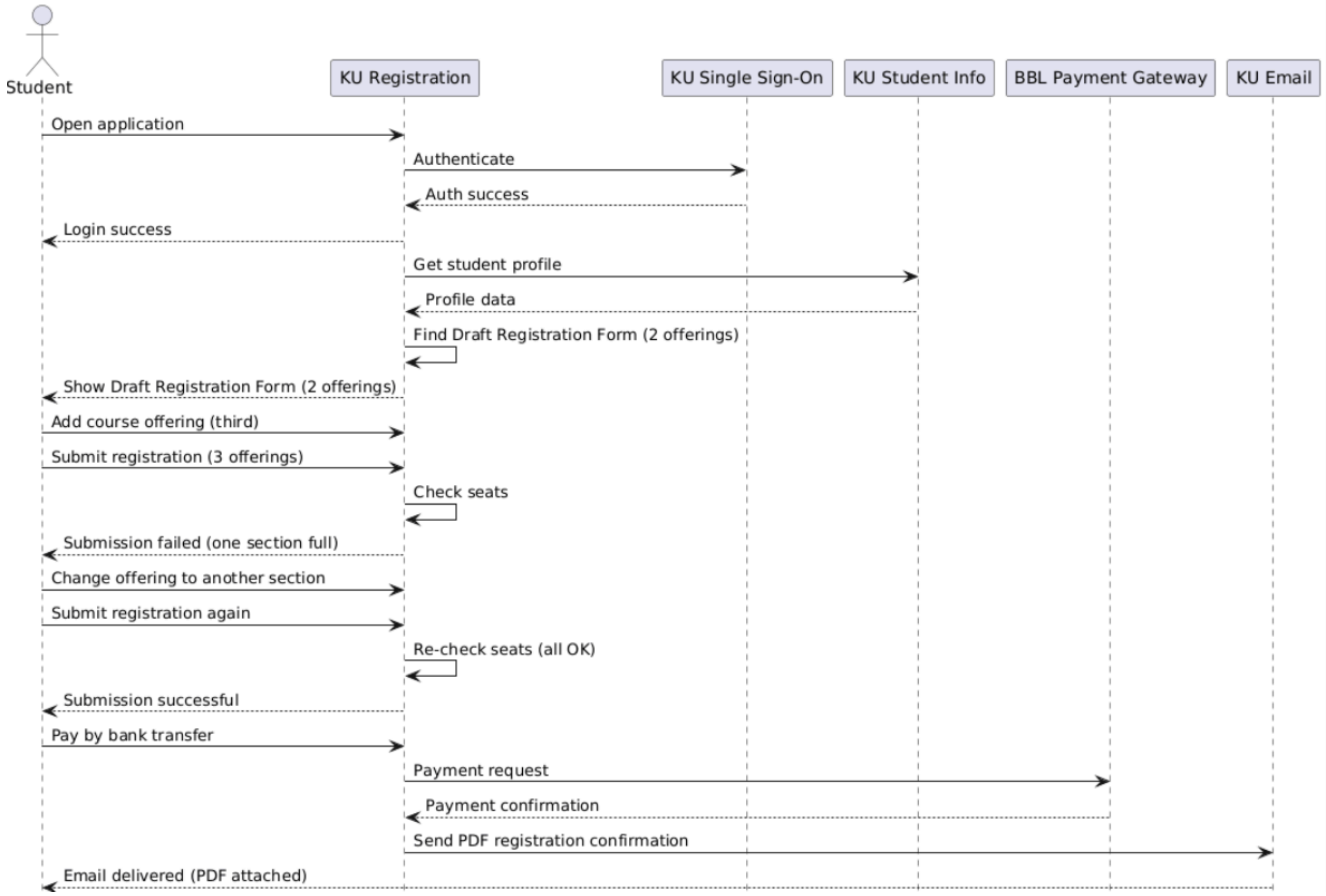
1.3 *10 points.* **Design a generic UI for the Student Registration Form View. Assume the semester is fixed to the current semester and student information is loaded from the logged-in user. Show required UI fields using leading "*", editable fields as textboxes, non-editable fields as striped textboxes as shown in the example below.**



1.4 *10 points.* **Draw the Class diagram for the data model used to store the Registration Form and its immediate (first-level) links to other classes. Do not include second-level links (i.e., anything more than one hop away from Registration Form's Header and Line Item classes). Show attributes, external calls, associations, multiplicities, inheritance, and composition. Do not show the class methods.**

Solutions:

Solution 1.1:



Solution 1.2:

Student Registration Form View **User**: Ying Kesuka **Login Name:** Yingke **Role**: Student

[Close][Save][Clear] (Form Controls)

Registration No.: _S2025-1-00232__ *Date: _15/09/2025__ [Date]

Username: _Yingke (from Login)__

Student ID: _(Logged-in Student Info)_ Student Name: _Ying Kesuka___

Registration Semester: _(from Active Semester)__ Total Credits: _15____

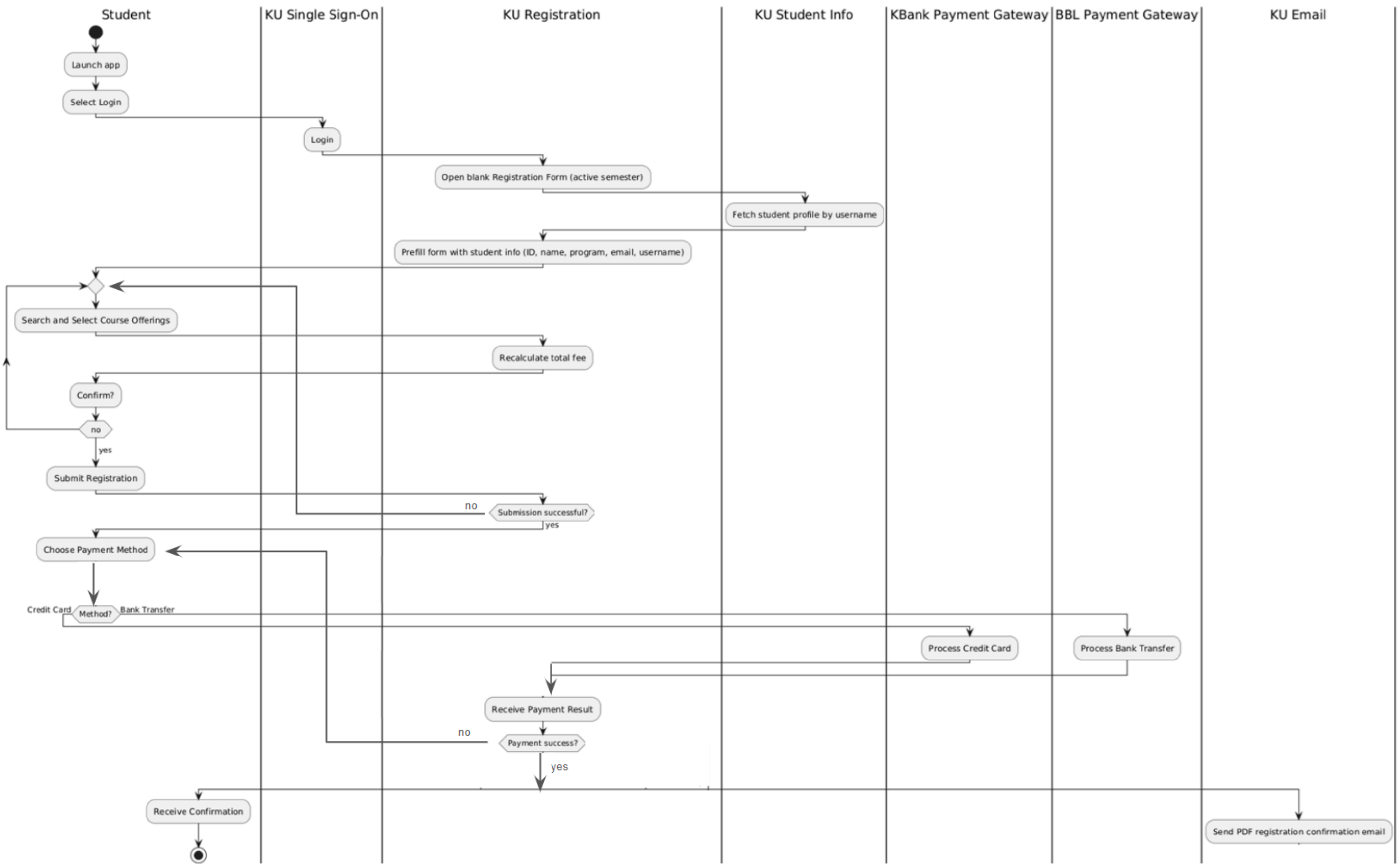
Total Price: _14,000.00_ Current Status: _Draft__ [SUBMIT]

Paid on Date: ________ (autoset at "Done") Paid By: ______(autoset after SUBMIT)

Enter Score and Grades for Students in this Course/Section:

#	*Course Offering ID	Course Code	Course Name	Semester	Section	Credits	Max Students	Price	Seats Remaining (view)
1	113 [LoV]	CPE334	Software Engineering	2025-1	1	3	40	8,000.00	3
2	119 [LoV]	MTH233	Calculus II	2025-1	7	4	40	6,000.00	12
3	[LoV]								
4	[LoV]								

Solution 1.3:



Solution 1.4:

Registration Form Data Model — 1st-Level Links Only